Metal2Humanoid

Building a Humanoid Robot from the Ground Up

MIT IAP 2027 | January 4-29, 2027 | Non-credit | Proposed syllabus

Course overview

A hands-on introduction to humanoid robotics, tracing the path from motors and actuators to coordinated motion and learned behavior. Over four weeks, students work in teams to assemble and test components of an open-source humanoid based on Berkeley Humanoid Lite. Topics include motors, transmissions, encoders, robot kinematics, feedback control, simulation, reinforcement learning, and transferring policies to physical hardware.

Each week combines two lectures, a guided hardware lab, and a scaffolded Python assignment. Projects progress from actuator position control to limb motion, simulated locomotion, and system integration, culminating in a team demonstration and brief technical report. Hardware components and software scaffolds are prepared in advance.

Teaching team

Ben Zandonati - Course proposer and lead instructor, MIT EECS
bzando@mit.edu

Willie McClinton - Co-instructor; curriculum and robotics instruction
Adjunct (Visiting) Faculty, University of Miami ECE; MIT alumnus (2026)
wbm3@mit.edu

Seth Robles - Co-instructor / undergraduate laboratory teaching assistant, MIT
sirobles@mit.edu

James Kim - Co-instructor / hardware laboratory instructor
Independent robotics consultant; Berkeley alumnus | heb3515@gmail.com

Faculty sponsor: Tomas Lozano-Perez, MIT EECS

Audience and prerequisites

Open to undergraduate and graduate students; estimated enrollment: 20. Introduction to computer science or robotics, or equivalent experience, is expected. Assignments use Python. No prior hardware assembly, control theory, or machine learning experience is required.

Learning objectives

- Explain how motors, transmissions, encoders, and feedback control produce controlled joint motion.
- Model robot limbs and implement basic end-effector motion in simulation.
- Train and evaluate a scaffolded locomotion policy and identify simulation-to-reality challenges.
- Integrate and test robot subsystems, document results, and demonstrate coordinated behavior.

Schedule and weekly activities

Lectures: Tuesdays and Thursdays, 1:00-2:30 p.m., January 5-28 (8 sessions).
Hardware labs: Fridays, 1:00-2:30 p.m., January 8-29 (4 sessions).
Office hours: One optional hour weekly; time to be arranged.
Location: To be confirmed. Separate lecture and laboratory spaces are planned, with a makerspace sought for labs.

18 scheduled instructional hours, plus optional office hours and independent assignments. Meeting times remain subject to room and instructor availability.

Week 1 | January 4-8 | Motor to Actuator

Lectures: Motor fundamentals; transmissions, encoders, and feedback control.
Lab: Assemble and test an actuator module from prepared components.
Assignment: Implement and evaluate motor position control.

Week 2 | January 11-15 | Actuator to Motion

Lectures: Robot modeling, forward and inverse kinematics, and trajectory generation.
Lab: Assemble actuators into a limb and validate joint motion.
Assignment: Implement end-effector motion control in simulation.

Week 3 | January 18-22 | Motion to Learning

Lectures: Reinforcement learning for locomotion; simulation-to-reality transfer.
Lab: Integrate limbs into the lower-body assembly and perform supported tests.
Assignment: Train and evaluate a locomotion policy using a scaffolded simulation environment.

Week 4 | January 25-29 | Learning to Humanoid

Lectures: Skill representations, sequencing, and integrated robot behavior.
Lab: Complete system integration and present team demonstrations.
Assignment: Implement a simple skill-sequencing task in simulation and submit a brief technical report documenting hardware and software results.

Participation, feedback, and final demonstration

Activities include four weekly assignments, laboratory participation, and a final team demonstration and report. Instructors provide feedback on understanding, testing, and documentation. No academic credit or formal grade is awarded. Successful untethered walking and manipulation are stretch goals. Simulation-based activities support continued learning when hardware requires troubleshooting.

Materials and support

Students will use prepared hardware kits, scaffolded programming exercises, and shared team builds. The organizing team will arrange support through the lab, external sponsors, or parts donors; commitments and the final budget will be established before the course begins. No EECS funding is requested.